

Food Waste Reduction Platform: Robust System Design and Project Management

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I. IMPLEMENTATION STRATEGY

The Implementation Strategy translates the Workshop 2 technical design and the Workshop 3 experimental validation into a concrete, reproducible development and deployment plan. It follows the best-practice collaboration guidelines established in *TeamWork_CS.pdf* (feature-branch workflow, atomic commits, mandatory peer reviews, and GitHub Projects board) and maintains full traceability to the 17 design requirements and the four-layer architecture defined in Workshop 2.

A. Implementation Phases

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B. Implementation Phases

- Phase 1 – Core Implementation and Experimentation (Workshop 3)** Full-stack prototype development using the Workshop 2 technology stack (completed by 23 May 2026).
- Phase 2 – Simulation, Pilot, and Final MVP (Workshop 4)** Realistic simulations, short controlled pilot, and delivery of all communication artifacts by 30 May 2026.

C. Required Infrastructure and Technology Stack

The platform is fully containerized to guarantee reproducibility and scalability:

All components are defined in a single `docker-compose.yml` file stored in the repository root, enabling one-command local deployment for development and testing.

D. Human Resources and Team Allocation

The implementation is executed by the four-member cross-functional team using the RACI matrix defined in the Project Charter (Section 4):

- Luna A. Sandoval Rodríguez** (Project Lead) – Overall coordination, integration, and quality gate reviews.
- Nicolás Rodríguez Granados** (Backend Developer) – Matching algorithm, APIs, and message broker.

Table I: Required Infrastructure and Technology Stack

Layer	Component	Technology
Frontend	Mobile Application	React Native / Expo (Android & iOS)
Backend	REST APIs & Business Logic	NestJS (Node.js)
Data	Persistent Storage & Geospatial Queries	PostgreSQL + PostGIS
Concurrency	Caching & Queues	Redis + Message Broker (RabbitMQ/Redis Streams)
Notifications	Push Notifications	Firebase Cloud Messaging (FCM)
Infrastructure	Containerization & Orchestration	Docker + Docker Compose
Deployment	Cloud Hosting (pilot)	AWS / Google Cloud (free tier for academic use)
Monitoring	Analytics & Dashboards	Custom Analytics Layer (PostgreSQL views)

- Cristian C. Bonilla Lizarazo** (Frontend Developer) – Mobile UI/UX and API integration.
- Juan S. Bravo Rojas** (Database Specialist) – PostGIS schema, data integrity, and analytics layer.

Weekly 45–60 minute coordination meetings and mandatory peer code reviews on GitHub ensure knowledge sharing and early detection of integration issues.

E. Estimated Budget and Resources

As an academic project, the budget is calculated primarily in human-hours with minimal monetary cost:

All software components are open-source or free-tier, ensuring zero licensing costs.

F. Rollout Timeline

The rollout follows a controlled, incremental approach aligned with the course schedule:

The Gantt chart illustrating the complete timeline is presented in Fig. ?? (Section 4).

Table II: Estimated Budget (Workshop 3 + Workshop 4)

Resource	Description	Cost
Human Capital	4 engineers $\times$ 10 h/week $\times$ 8 remaining weeks	320 HH (no monetary cost)
Cloud Infrastructure	Docker hosting + PostgreSQL + FCM (free tier)	\$0 (academic tier)
Production Hosting	AWS/Google Cloud (estimated for 8-week pilot)	\$50 USD (theoretical)
Development Tools	GitHub, Docker, Expo, PostGIS	\$0 (open source)
Total Estimated Monetary Cost		\$50 USD

Table III: Rollout Timeline (Workshop 3 $\rightarrow$ Final MVP)

Milestone	Date	Deliverables
Core Prototype Complete	23 May 2026	Full implementation, unit/integration tests, risk register validation (Workshop 4 delivery)
Simulation	20–23 May 2026	300-user load tests, performance benchmarking, sensitivity analysis
Pilot Deployment	24–30 May 2026	Short controlled validation pilot within 3 km radius (Engineering Faculty area)
Final MVP	30 May 2026	Production-ready application + administrative dashboard
Communication Deliverables	30 May 2026	Scientific paper, academic poster, technical report, and final presentation

G. Resource Management Plan

1) *Resource Allocation and Tracking*: Resource usage is tracked through the GitHub Projects Kanban board and weekly burndown charts. Each sprint is limited to a maximum of 10 hours per team member to prevent burnout. Effort distribution is aligned with the RACI matrix defined in the Project Charter:

Table IV: Resource Allocation Overview (Workshop 3–4)

Resource Type	Management Mechanism	Planned Effort / Cost
Human Resources	GitHub Projects + weekly burndown charts	640 HH total (4 engineers $\times$ 10 h/week)
Technical Infrastructure	Docker Compose + free-tier cloud services	\$0 (academic tier)
Financial Resources	Monitored in project board	\$0–50 (theoretical pilot hosting)

2) Resource Monitoring and Control:

- **Weekly Resource Review**: During the 45–60 minute status meeting, the team reviews burndown charts and flags any deviation greater than 15%.
- **Automatic Alerts**: GitHub issue labels (`resource:critical`, `resource:overallocated`) trigger notifications to the Project Lead.
- **Control Gates**: No merge to `develop` is allowed without peer review and confirmation that the assigned effort was not exceeded.

- **Contingency**: If any member exceeds 12 hours in a week, the Project Lead reassigns tasks immediately to maintain balance.

This plan guarantees that the project remains within academic constraints, ensures full traceability of effort, and supports the successful delivery of the final MVP and communication artifacts by **30 May 2026**.

REFERENCES

- [1] Workshop 1 – Food Waste Reduction Mobile Application, A Rigorous System Analysis, Universidad Distrital Francisco José de Caldas, 2026.
- [2] Workshop 2 – System Design Document, Universidad Distrital Francisco José de Caldas, 2026.
- [3] IEEE Standards Association, “IEEE 730-2014 - IEEE Standard for Software Quality Assurance Processes,” 2014.